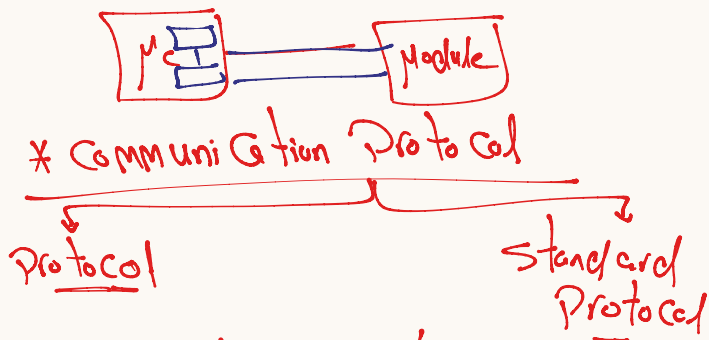
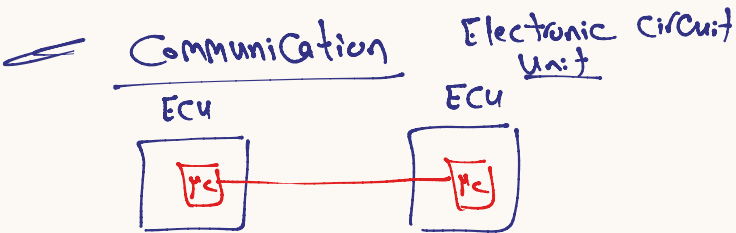
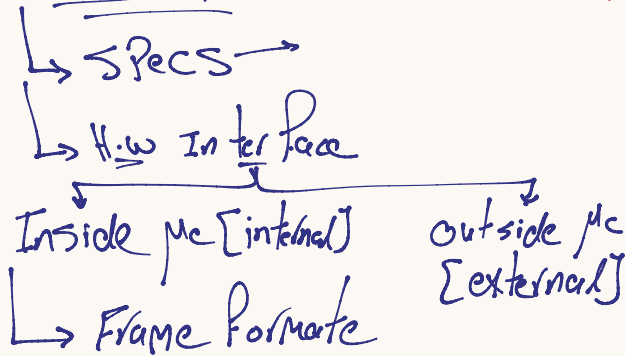
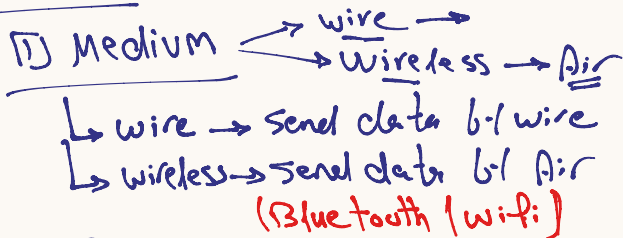




* any Protocol or standard Protocol



* Specs

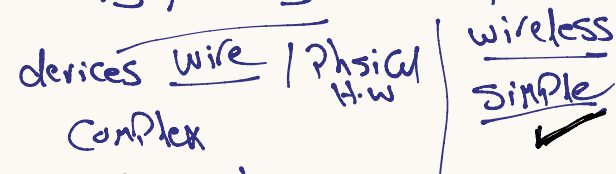


→ Compare

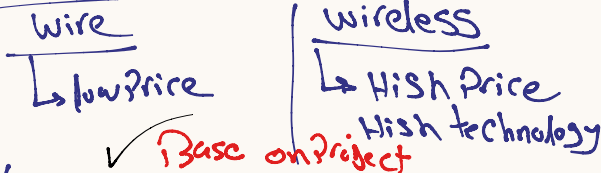
1) Security



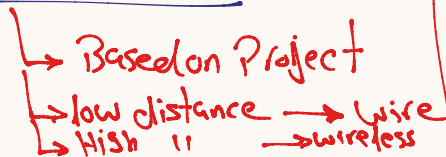
2) Mobility



3) Cost



4) distance



5) speed

wire faster than wireless

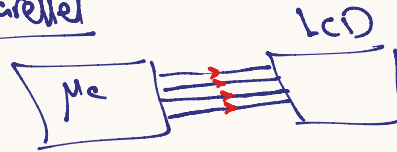
2) transmission technique

Serial Parallel

* Serial :: bit b/w bit ✓ in Embedded

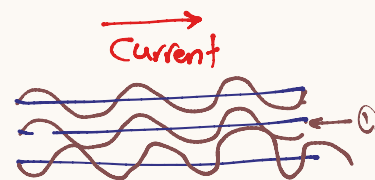
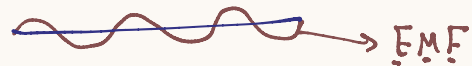


* Parallel

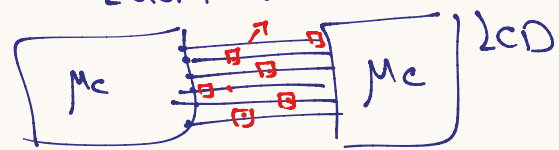


* Parallel Problems

- 1) Complex Connection
- 2) crosstalk (Magnetic Field Interference)



3) Clock skew [delay data transmission]



3) Receiver / transmitter relation

Peer to Peer / Node to Node

2 MCU only



Master / slave

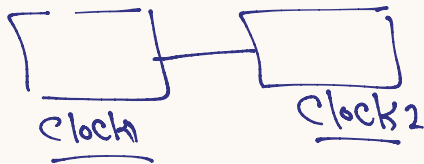
* Master → able to Init Communicate.

* slave → Can't Init Communicate

type Init Communicate How

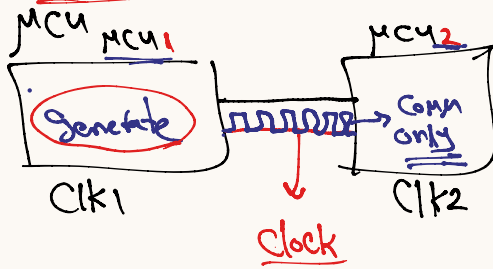
- 1) Single Master, Multi slaves.
- 2) Multi Master, Multi slaves.
- 3) Multi Master, No Slaves.

[4] Synchronization / Asynchronization

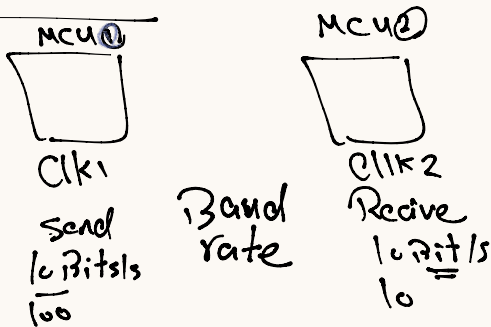


* Synchronization

↳ shared clock wire between



* Asynchronization



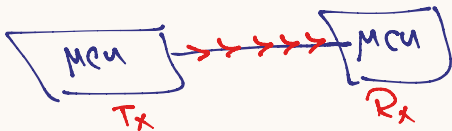
Band rate!.. Symbols in sec.

bit. byte word Pages

Common

[5] Data Direction

↳ ① Simplex!.. Radio / TV



↳ ② Half duplex!.. Hoki to ki



Can Tx & Rx but Not Same time

↳ ③ Full duplex!.. Rx & Tx at same time

[6] Throughput

Ration between No of Data and No of frame.

$$\begin{aligned} &\text{frame} \rightarrow 50 \rightarrow \text{Data} \rightarrow 45 \\ &\text{Throughput} = \frac{45}{50} \times 100 = 90\% \end{aligned}$$

$$\begin{aligned} &\text{frame} \rightarrow 3h \rightarrow \text{Data} \rightarrow 30s \\ &\text{Throughput} \rightarrow \frac{30s}{3 \times 60 \times 60} \times 100 \end{aligned}$$

uart → Specs

Power Bug